

# ASIC Physical Design Complete Explanations

Every module explained with what it is, why it matters, examples, common mistakes and practice questions

| Level        | Modules Covered                                             |
|--------------|-------------------------------------------------------------|
| Beginner     | Digital fundamentals, ASIC flow, HDL/Verilog/VHDL           |
| Intermediate | RTL to gate netlist, technology files, STA, PD intro        |
| Advanced     | Floorplan, power plan, metal stack, placement, CTS, routing |
| Expert       | Signoff, DRC/LVS, IR/EM, SI, ECO and tapeout readiness      |

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# How To Use This PDF

This version is designed as learning notes, not just a syllabus. Each topic is explained in a beginner-friendly way and connected to real ASIC physical design work. Read one topic at a time, copy the examples into your notebook, and answer the practice questions before moving forward.

| Section         | Meaning                                                        |
|-----------------|----------------------------------------------------------------|
| What it is      | Plain-language definition of the concept.                      |
| Why it matters  | How the concept appears in ASIC, PnR, STA, signoff or tapeout. |
| Examples        | Small practical examples so the idea becomes concrete.         |
| Common mistakes | Typical beginner errors to avoid.                              |
| Practice        | Questions to test whether you really understood the topic.     |

Recommended study method: spend 30-45 minutes per topic. First read the definition, then draw one diagram, then explain the topic aloud as if you are in an interview.

# BEGINNER - Digital Fundamentals for ASIC Physical Design

This module builds the base used everywhere in ASIC design: binary data, logic gates, combinational logic, sequential logic, clocks and reset.

| # | Topic                             | Main Skill                                                                                         |
|---|-----------------------------------|----------------------------------------------------------------------------------------------------|
| 1 | Number Systems                    | RTL, waveforms, register maps, memory addresses, scan data and configuration values are usually... |
| 2 | Boolean Algebra and Logic Gates   | Synthesis converts RTL into gates. Physical design sees instances such as NAND2_X1, INV_X4 and ... |
| 3 | Combinational Circuits            | Most timing paths between flops pass through combinational logic. More logic depth usually mean... |
| 4 | Sequential Circuits               | ASIC timing is mostly checked from one sequential element to another. Placement, CTS and STA ar... |
| 5 | Clock, Reset and Timing Intuition | Every physical design stage protects timing. CTS controls clock arrival; placement and routing ... |

## Number Systems

| Part                      | Explanation                                                                                                                                                                                                                    |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What it is                | A number system is a way to represent values using symbols. In digital design, hardware stores and processes values as binary bits, but engineers often read the same values in decimal, hexadecimal or octal for convenience. |
| Why it matters in ASIC/PD | RTL, waveforms, register maps, memory addresses, scan data and configuration values are usually shown in binary or hexadecimal. If you cannot convert between them, debug becomes slow.                                        |

## Examples

- Binary 1010 means decimal 10 because  $1 \times 8 + 0 \times 4 + 1 \times 2 + 0 \times 1 = 10$ .
- Hex A is binary 1010. Hex 3F is binary 0011 1111 and decimal 63.
- An 8-bit bus can represent 0 to 255 unsigned values.

## Common Mistakes

- Thinking 10 in binary equals decimal ten; binary 10 is decimal two.
- Forgetting bit width. Hex F can be 4'b1111, 8'b00001111 or wider depending on context.

## Practice Questions

- Convert decimal 45 to binary and hex.
- Write 8'hA5 in binary.
- What is the maximum unsigned value of a 10-bit bus?

## Boolean Algebra and Logic Gates

| Part                      | Explanation                                                                                                                                                                 |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What it is                | Boolean algebra handles values 0 and 1 using operations such as AND, OR, NOT, NAND, NOR and XOR. Gates are the physical logic cells that implement these operations.        |
| Why it matters in ASIC/PD | Synthesis converts RTL into gates. Physical design sees instances such as NAND2_X1, INV_X4 and DFF_X1, so knowing the logic helps you understand netlists and timing paths. |

### Examples

- A AND B is 1 only when both A and B are 1.
- A OR B is 1 when at least one input is 1.
- XOR is useful for adders because it detects when inputs differ.

### Common Mistakes

- Memorizing truth tables without understanding behavior.
- Forgetting NAND and NOR are common because they are efficient in CMOS.

### Practice Questions

- Draw the truth table for XOR.
- Implement NOT using NAND.
- Which gate output toggles when only one input changes?

## Combinational Circuits

| Part                      | Explanation                                                                                                                                      |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| What it is                | Combinational logic output depends only on current inputs. Examples include muxes, decoders, encoders, adders, comparators and arithmetic logic. |
| Why it matters in ASIC/PD | Most timing paths between flops pass through combinational logic. More logic depth usually means more delay and harder setup timing.             |

## Examples

- A 2:1 mux selects between input A and B using select S.
- A decoder converts binary input into one-hot output.
- A ripple-carry adder is simple but can be slow because carry propagates bit by bit.

## Common Mistakes

- Creating incomplete conditions in RTL, which can infer latches.
- Ignoring logic depth in critical paths.

## Practice Questions

- Draw a 4:1 mux from 2:1 muxes.
- Explain why a long adder can hurt setup timing.
- Where would a decoder appear in a memory system?

## Sequential Circuits

| Part                      | Explanation                                                                                                                                                              |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What it is                | Sequential logic stores state. Flip-flops sample data on a clock edge. Latches are level-sensitive storage elements. FSMs use flops plus combinational next-state logic. |
| Why it matters in ASIC/PD | ASIC timing is mostly checked from one sequential element to another. Placement, CTS and STA are built around flop-to-flop timing paths.                                 |

## Examples

- A D flip-flop captures D at the rising edge of clock and outputs Q.
- A counter increments its stored value every clock.
- An FSM moves between states such as IDLE, READ and WRITE.

## Common Mistakes

- Using latches accidentally through incomplete RTL.
- Ignoring asynchronous reset release timing.

## Practice Questions

- Draw a flop-to-flop timing path.
- Explain the difference between latch and flop.
- What happens if reset fanout is extremely high?

## Clock, Reset and Timing Intuition

| Part                      | Explanation                                                                                                                                                                       |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What it is                | Clock controls when flops capture data. Reset initializes state. Setup time means data must arrive before the clock edge; hold time means data must remain stable after the edge. |
| Why it matters in ASIC/PD | Every physical design stage protects timing. CTS controls clock arrival; placement and routing control data delay.                                                                |

## Examples

- If data arrives late, setup violation occurs.
- If data changes too early after the capture edge, hold violation occurs.
- A high-fanout reset can create routing and transition issues.

## Common Mistakes

- Thinking faster data is always better; too-fast data can cause hold violations.
- Treating reset as a normal signal without checking fanout and recovery/removal.

## Practice Questions

- Define setup and hold in one sentence each.
- Why can buffer insertion fix hold?
- Why does CTS change timing reports?



# BEGINNER - ASIC Flow Awareness and Architecture to Signoff

This module explains the full chip journey: architecture, RTL, verification, synthesis, DFT, PnR, STA, physical verification and tapeout.

| # | Topic                              | Main Skill                                                                                         |
|---|------------------------------------|----------------------------------------------------------------------------------------------------|
| 1 | Architecture and Microarchitecture | Physical design quality depends on architectural choices such as memory placement, clock domain... |
| 2 | Frontend vs Backend                | A physical design engineer must communicate with frontend teams when constraints, clock definit... |
| 3 | Synthesis to PnR Handoff           | Bad handoff causes wasted implementation time. Missing clocks, wrong libraries or unresolved ce... |
| 4 | Signoff Meaning                    | Tapeout is expensive. A design must be proven manufacturable and functionally safe before sendi... |

## Architecture and Microarchitecture

| Part                      | Explanation                                                                                                                                                              |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What it is                | Architecture defines what the chip must do. Microarchitecture defines how the design is organized internally using blocks, pipelines, buses, memories and control logic. |
| Why it matters in ASIC/PD | Physical design quality depends on architectural choices such as memory placement, clock domains, throughput, power modes and block partitioning.                        |

## Examples

- A processor architecture may require instruction fetch, decode and execute.
- A memory-heavy design may need SRAM macros close to compute logic.
- A pipeline improves frequency but adds flops and clock power.

## Common Mistakes

- Thinking physical design starts only after synthesis; many PD risks are created by architecture.
- Ignoring macro count and clock-domain count early.

## Practice Questions

- List three architecture choices that affect floorplan.
- Why do SRAM macros matter in PnR?
- How does pipelining affect timing?

## Frontend vs Backend

| Part                      | Explanation                                                                                                                                        |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| What it is                | Frontend focuses on RTL, verification, synthesis and DFT. Backend focuses on physical implementation, timing closure, routing and signoff.         |
| Why it matters in ASIC/PD | A physical design engineer must communicate with frontend teams when constraints, clock definitions, scan chains or netlist issues affect closure. |

## Examples

- Frontend delivers RTL, SDC and synthesized netlist.
- Backend delivers DEF, GDS, SPEF, timing reports and signoff package.
- DFT delivers scan information that affects placement and routing.

## Common Mistakes

- Treating SDC as perfect without review.
- Not asking design owners about suspicious false paths.

## Practice Questions

- Name three frontend deliverables.
- Name three backend deliverables.
- Who owns scan DEF and why does PD care?

## Synthesis to PnR Handoff

| Part                      | Explanation                                                                                                                                                  |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What it is                | Synthesis converts RTL into a gate-level netlist using timing constraints and standard-cell libraries. PnR takes this netlist and creates a physical layout. |
| Why it matters in ASIC/PD | Bad handoff causes wasted implementation time. Missing clocks, wrong libraries or unresolved cells can block floorplan and placement.                        |

## Examples

- Netlist.v contains cells such as NAND2\_X1 and DFF\_X1.
- SDC contains create\_clock and set\_input\_delay.
- Area and timing reports show whether synthesis is reasonable before PnR.

## Common Mistakes

- Starting PnR before checking unresolved references.
- Using netlist and SDC from different synthesis runs.

## Practice Questions

- What files are required before PnR can start?
- Why must .lib and LEF cell names match?
- What is an unresolved reference?

## Signoff Meaning

| Part                      | Explanation                                                                                                                                   |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| What it is                | Signoff means the design passed required final checks: timing, power integrity, physical verification, reliability and release documentation. |
| Why it matters in ASIC/PD | Tapeout is expensive. A design must be proven manufacturable and functionally safe before sending GDS/OASIS to the foundry.                   |

## Examples

- STA clean means setup and hold pass across scenarios.
- DRC clean means layout obeys foundry rules.
- LVS clean means layout matches netlist connectivity.

## Common Mistakes

- Thinking routed DEF alone is tapeout-ready.
- Ignoring waivers and report ownership.

## Practice Questions

- Define DRC and LVS.
- What does WNS  $\geq 0$  mean?
- Why are checksums used in release packages?

# BEGINNER - HDL, VHDL and Verilog Fundamentals

This module teaches enough HDL to read design intent, understand synthesis results and debug physical-design issues caused by RTL style.

| # | Topic                    | Main Skill                                                                                         |
|---|--------------------------|----------------------------------------------------------------------------------------------------|
| 1 | Verilog Module Basics    | Physical designers often inspect RTL or netlists to understand clocks, resets, hierarchy and cr... |
| 2 | Blocking vs Non-Blocking | Wrong assignment style can create mismatches between intended hardware and simulation, causing ... |
| 3 | VHDL Basics              | Some projects mix Verilog/SystemVerilog with VHDL IP. Backend engineers should recognize hierar... |
| 4 | Synthesizable Coding     | Non-synthesizable RTL blocks design implementation. Poor coding style can infer latches, huge m... |

## Verilog Module Basics

| Part                      | Explanation                                                                                                                                          |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| What it is                | A Verilog module describes hardware using ports, wires, registers and statements. It is not software; synthesizable Verilog becomes gates and flops. |
| Why it matters in ASIC/PD | Physical designers often inspect RTL or netlists to understand clocks, resets, hierarchy and critical paths.                                         |

## Examples

- module counter(input clk, input rst\_n, output reg [3:0] q); describes a 4-bit counter interface.
- assign y = a & b; becomes an AND gate.
- always @(posedge clk) q <= d; becomes a flop.

## Common Mistakes

- Reading Verilog as sequential software.
- Using unsynthesizable constructs for hardware logic.

## Practice Questions

- Write a 2:1 mux in Verilog.
- Which line creates a flop?
- What does output [7:0] q mean?

## Blocking vs Non-Blocking

| Part                      | Explanation                                                                                                                           |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| What it is                | Blocking assignment (=) updates immediately in simulation order. Non-blocking assignment (<=) schedules updates for clocked behavior. |
| Why it matters in ASIC/PD | Wrong assignment style can create mismatches between intended hardware and simulation, causing late functional or timing debug.       |

## Examples

- Use <= in clocked always blocks.
- Use = in combinational always blocks when assigning temporary logic.
- q <= d inside posedge block infers a flop.

## Common Mistakes

- Using blocking assignments in sequential logic without understanding ordering.
- Mixing styles randomly in one block.

## Practice Questions

- Which assignment should be used for flops?
- Why can blocking cause simulation surprises?
- Write a safe clocked always block.



## VHDL Basics

| Part                      | Explanation                                                                                                                                                    |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What it is                | VHDL describes hardware using entity, architecture, signals and processes. It is strongly typed and widely used in aerospace, FPGA and some ASIC environments. |
| Why it matters in ASIC/PD | Some projects mix Verilog/SystemVerilog with VHDL IP. Backend engineers should recognize hierarchy and ports across languages.                                 |

## Examples

- entity defines ports.
- architecture defines internal behavior.
- process(clk) with rising\_edge(clk) describes sequential logic.

## Common Mistakes

- Confusing signal assignment timing with software assignment.
- Ignoring type conversions in arithmetic.

## Practice Questions

- What is the VHDL equivalent of a Verilog module interface?
- What does rising\_edge(clk) imply?
- Where are ports declared?

## Synthesizable Coding

| Part                      | Explanation                                                                                                                             |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| What it is                | Synthesizable code can be converted to gates and flops. Testbench-only code, delays and file operations usually cannot become hardware. |
| Why it matters in ASIC/PD | Non-synthesizable RTL blocks design implementation. Poor coding style can infer latches, huge muxes or bad timing paths.                |

## Examples

- always\_ff @(posedge clk) is sequential.
- always\_comb with full assignments avoids latches.
- #10 delay is simulation-only.

## Common Mistakes

- Incomplete if/case causing latch inference.
- Huge combinational logic in one cycle causing timing failure.

## Practice Questions

- Find latch risk in an incomplete case statement.
- Why are # delays not synthesizable?
- What is the hardware for if-else priority logic?

# INTERMEDIATE - RTL to Gate-Level Netlist and Technology Fundamentals

This module connects HDL to real technology files used by synthesis and physical design.

| # | Topic                            | Main Skill                                                                                         |
|---|----------------------------------|----------------------------------------------------------------------------------------------------|
| 1 | Gate-Level Netlist               | PnR places and routes this netlist. Timing, area and power closure happen on actual technology ... |
| 2 | Standard-Cell Libraries          | Physical design tools need timing views for STA and physical views for placement/routing.          |
| 3 | Technology LEF, ITF, TLU+ and RC | Routing and timing depend heavily on wire resistance and capacitance. Wrong RC technology gives... |
| 4 | SDC Constraints                  | Tools optimize only what constraints describe. Bad constraints can hide real violations or crea... |

## Gate-Level Netlist

| Part                      | Explanation                                                                                                                        |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| What it is                | A gate-level netlist is a structural Verilog file containing standard-cell instances and connectivity after synthesis.             |
| Why it matters in ASIC/PD | PnR places and routes this netlist. Timing, area and power closure happen on actual technology cells, not abstract RTL statements. |

## Examples

- U1 NAND2\_X1 (.A(a), .B(b), .ZN(n1)); is a cell instance.
- DFF\_X1 cells represent sequential state.
- BUF\_X4 may be inserted to drive high fanout.

## Common Mistakes

- Expecting the netlist to look like RTL.
- Editing netlist manually without controlled ECO flow.

## Practice Questions

- Identify cell name, instance name and nets in a netlist line.
- What is a buffer used for?
- Why does synthesis choose different drive strengths?

## Standard-Cell Libraries

| Part                      | Explanation                                                                                                                                                                  |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What it is                | A standard-cell library is a collection of cells such as INV, NAND, NOR, flop, latch, tie, filler and clock buffers characterized for timing, power and physical dimensions. |
| Why it matters in ASIC/PD | Physical design tools need timing views for STA and physical views for placement/routing.                                                                                    |

## Examples

- .lib has timing arcs and power.
- .db is compiled Synopsys library format.
- LEF has size, pins and obstructions.

## Common Mistakes

- Mixing Liberty and LEF from different library versions.
- Using non-clock buffers in CTS.

## Practice Questions

- Difference between .lib and LEF?
- What does X1/X2/X4 mean?
- Why are clock cells special?

## Technology LEF, ITF, TLU+ and RC

| Part                      | Explanation                                                                                                                                                      |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What it is                | Technology files describe routing layers, via rules, width, spacing, pitch and extraction models. They teach tools how wires behave physically and electrically. |
| Why it matters in ASIC/PD | Routing and timing depend heavily on wire resistance and capacitance. Wrong RC technology gives wrong timing results.                                            |

### Examples

- Tech LEF defines M1, M2, via12 and layer directions.
- ITF describes interconnect stack for extraction.
- TLU+ or QRC tech is used for delay calculation.

### Common Mistakes

- Using wrong RC corner for setup/hold.
- Ignoring layer preferred directions.

### Practice Questions

- What file defines metal pitch?
- Why do top metals have lower resistance?
- What is a via enclosure rule?

## SDC Constraints

| Part                      | Explanation                                                                                                      |
|---------------------------|------------------------------------------------------------------------------------------------------------------|
| What it is                | SDC tells tools about clocks, IO timing, uncertainties, generated clocks, false paths and multicycle paths.      |
| Why it matters in ASIC/PD | Tools optimize only what constraints describe. Bad constraints can hide real violations or create fake problems. |

## Examples

- `create_clock -period 2.0 [get_ports clk]` defines 500 MHz clock.
- `set_input_delay` models external launch timing.
- `set_false_path` excludes paths that should not be timed.

## Common Mistakes

- Overusing false paths.
- Missing generated clocks.
- No unconstrained path check.

## Practice Questions

- Write a 1 GHz `create_clock`.
- What is multicycle path?
- Why is an unconstrained endpoint dangerous?

## INTERMEDIATE - Static Timing Analysis

This module explains how ASIC teams prove timing without simulating every possible vector.

| # | Topic                       | Main Skill                                                                         |
|---|-----------------------------|------------------------------------------------------------------------------------|
| 1 | Setup Timing                | Setup failures mean the chip may not run at target frequency.                      |
| 2 | Hold Timing                 | Hold failures are frequency-independent and must be fixed for silicon correctness. |
| 3 | WNS, TNS and Timing Reports | Engineers use these reports to prioritize timing closure.                          |
| 4 | Corners and OCV             | Silicon must work under manufacturing and operating variation.                     |



## Setup Timing

| Part                      | Explanation                                                                                                          |
|---------------------------|----------------------------------------------------------------------------------------------------------------------|
| What it is                | Setup timing checks that data launched by one clock edge arrives at the capture flop before the required setup time. |
| Why it matters in ASIC/PD | Setup failures mean the chip may not run at target frequency.                                                        |

## Examples

- If clock period is 2 ns and data path needs 2.2 ns, setup violation occurs.
- Buffering, resizing or pipelining can improve setup.
- Slow corner is often setup-critical.

## Common Mistakes

- Fixing setup by blindly adding buffers.
- Ignoring clock uncertainty.

## Practice Questions

- What does negative setup slack mean?
- How can placement improve setup?
- Why does high capacitance hurt setup?

## Hold Timing

| Part                      | Explanation                                                                                             |
|---------------------------|---------------------------------------------------------------------------------------------------------|
| What it is                | Hold timing checks that data does not arrive too early at the capture flop after the active clock edge. |
| Why it matters in ASIC/PD | Hold failures are frequency-independent and must be fixed for silicon correctness.                      |

## Examples

- Adding delay buffers can fix hold.
- Fast corner is often hold-critical.
- CTS can create new hold violations by changing skew.

## Common Mistakes

- Thinking lowering frequency fixes hold.
- Fixing hold so aggressively that setup breaks.

## Practice Questions

- Why is hold frequency-independent?
- How does useful skew affect hold?
- Name two hold-fix methods.

## WNS, TNS and Timing Reports

| Part                      | Explanation                                                                                                                                                   |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What it is                | WNS is worst negative slack. TNS is total negative slack across violating paths. Timing reports show path startpoint, endpoint, clock, cells, nets and slack. |
| Why it matters in ASIC/PD | Engineers use these reports to prioritize timing closure.                                                                                                     |

## Examples

- WNS = -0.120 ns means worst path misses by 120 ps.
- TNS = 0 means no accumulated negative slack.
- A path report can show whether delay is cell-dominated or net-dominated.

## Common Mistakes

- Looking only at WNS and ignoring many small violations.
- Ignoring unconstrained paths.

## Practice Questions

- Which is worse: WNS -10 ps with TNS -500 ns, or WNS -200 ps with TNS -200 ps?
- What is a startpoint?
- What is an endpoint?

## Corners and OCV

| Part                      | Explanation                                                                                                         |
|---------------------------|---------------------------------------------------------------------------------------------------------------------|
| What it is                | Timing is checked across process, voltage, temperature and RC corners. OCV/AOCV/POCV model variation between paths. |
| Why it matters in ASIC/PD | Silicon must work under manufacturing and operating variation.                                                      |

## Examples

- Slow process, low voltage, high temperature is setup-critical.
- Fast process, high voltage, low temperature is hold-critical.
- Cmax RC increases wire delay.

## Common Mistakes

- Closing timing in only one corner.
- Using setup corner for hold signoff.

## Practice Questions

- Which corner is usually setup-critical?
- What is derating?
- Why does voltage affect delay?

# INTERMEDIATE - Physical Design Introduction

This module introduces backend implementation concepts before going deep into floorplan, placement, CTS and routing.

| # | Topic                       | Main Skill                                                                         |
|---|-----------------------------|------------------------------------------------------------------------------------|
| 1 | DEF, GDS, SPEF and SDF      | These are core files used in implementation, verification, simulation and tapeout. |
| 2 | Rows, Sites and Tracks      | Placement legality and routing access depend on proper row/site/track alignment.   |
| 3 | Macros, Halos and Blockages | Macro placement strongly affects congestion, timing and power grid quality.        |

## DEF, GDS, SPEF and SDF

| Part                      | Explanation                                                                                                                              |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| What it is                | DEF stores physical placement/routing information. GDS/OASIS stores final mask layout. SPEF stores parasitics. SDF stores timing delays. |
| Why it matters in ASIC/PD | These are core files used in implementation, verification, simulation and tapeout.                                                       |

## Examples

- DEF can show cell locations and nets.
- GDS is sent to foundry.
- SPEF feeds PrimeTime/Tempus signoff STA.

## Common Mistakes

- Confusing DEF with final tapeout layout.
- Running final STA without extracted SPEF.

## Practice Questions

- What is SPEF used for?
- Which file goes to foundry?
- What does DEF contain?

## Rows, Sites and Tracks

| Part                      | Explanation                                                                                                |
|---------------------------|------------------------------------------------------------------------------------------------------------|
| What it is                | Rows and sites define legal standard-cell placement. Tracks define routing grid positions for metal wires. |
| Why it matters in ASIC/PD | Placement legality and routing access depend on proper row/site/track alignment.                           |

## Examples

- A 9-track cell has height based on routing tracks.
- Cells are placed in rows with alternating orientation.
- M1/M2 tracks must align with pins.

## Common Mistakes

- Ignoring mixed-height cells.
- Creating floorplan with poor pin access.

## Practice Questions

- What is a placement site?
- Why alternate row orientation?
- How do tracks affect routability?

## Macros, Halos and Blockages

| Part                      | Explanation                                                                                                              |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------|
| What it is                | Macros are large pre-designed blocks such as SRAM, ROM, PLL or analog IP. Halos and blockages reserve space around them. |
| Why it matters in ASIC/PD | Macro placement strongly affects congestion, timing and power grid quality.                                              |

### Examples

- SRAM pins should face standard-cell logic.
- A hard blockage prevents cells in a region.
- A routing blockage can reserve metal layers around macros.

### Common Mistakes

- Macros too close together.
- No halo near pin-dense macro edge.

### Practice Questions

- Why place macros near boundary?
- What is a halo?
- When use partial blockage?



# ADVANCED - Floorplanning and Power Planning

This module teaches the first major backend stage: creating a routable, timing-friendly and power-safe chip/block canvas.

| # | Topic                              | Main Skill                                                                                         |
|---|------------------------------------|----------------------------------------------------------------------------------------------------|
| 1 | Die, Core and Utilization          | Too-high utilization causes congestion and timing closure pain. Too-low utilization wastes area... |
| 2 | Macro and Port Placement           | Good macro/port placement reduces wirelength, congestion and timing criticality.                   |
| 3 | Power Grid                         | Weak PG causes IR drop, EM violations and possible silicon failure.                                |
| 4 | Tap, Endcap, Spare and Decap Cells | A design can be timing clean but fail DRC/latch-up/reliability rules without these cells.          |

## Die, Core and Utilization

| Part                      | Explanation                                                                                                                |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------|
| What it is                | Die is total boundary; core is standard-cell placement area. Utilization is used cell area divided by available core area. |
| Why it matters in ASIC/PD | Too-high utilization causes congestion and timing closure pain. Too-low utilization wastes area.                           |

## Examples

- 70% utilization means 30% whitespace for routing and optimization.
- Macros reduce available standard-cell area.
- Aspect ratio affects path lengths.

## Common Mistakes

- Starting with 85% utilization for a routing-heavy design.
- Ignoring future CTS/hold buffer insertion.

## Practice Questions

- Why leave whitespace?
- What happens if utilization is too high?
- How does aspect ratio affect timing?

## Macro and Port Placement

| Part                      | Explanation                                                                                                        |
|---------------------------|--------------------------------------------------------------------------------------------------------------------|
| What it is                | Macro placement decides where large IP blocks sit. Port placement decides where signals enter and leave the block. |
| Why it matters in ASIC/PD | Good macro/port placement reduces wirelength, congestion and timing criticality.                                   |

## Examples

- Place connected macros close.
- Put ports near related logic.
- Keep analog/PLL away from noisy switching regions.

## Common Mistakes

- Pin sides facing away from logic.
- Random port order causing crisscross routes.

## Practice Questions

- What is a macro channel?
- Why group interface pins?
- How does macro orientation matter?

## Power Grid

| Part                      | Explanation                                                               |
|---------------------------|---------------------------------------------------------------------------|
| What it is                | Power grid distributes VDD and VSS through rings, straps, rails and vias. |
| Why it matters in ASIC/PD | Weak PG causes IR drop, EM violations and possible silicon failure.       |

## Examples

- Top metals carry long-distance power because they are thick and low resistance.
- Followpin rails feed standard cells.
- Via arrays connect straps between layers.

## Common Mistakes

- Unconnected macro power pins.
- Power mesh too dense, blocking signal routes.

## Practice Questions

- Why use top metals for power?
- What is IR drop?
- What is sroute?

## Tap, Endcap, Spare and Decap Cells

| Part                      | Explanation                                                                                                  |
|---------------------------|--------------------------------------------------------------------------------------------------------------|
| What it is                | Physical-only cells support manufacturing, reliability and ECO flexibility. They do not implement RTL logic. |
| Why it matters in ASIC/PD | A design can be timing clean but fail DRC/latch-up/reliability rules without these cells.                    |

### Examples

- Tap cells prevent latch-up.
- Endcaps protect row edges.
- Decaps reduce dynamic supply noise.
- Spare cells help metal ECOs.

### Common Mistakes

- Forgetting filler after ECO.
- Clustering all spare cells in one corner.

### Practice Questions

- Why use tap cells?
- What does decap do?
- What is a spare cell used for?

# ADVANCED - Metal Stack, Placement, CTS and Routing

This module covers the core implementation work where standard cells, clocks and wires become a legal physical layout.

| # | Topic                                       | Main Skill                                                                                         |
|---|---------------------------------------------|----------------------------------------------------------------------------------------------------|
| 1 | Metal Stack, Width, Spacing, Pitch and Vias | Routing quality, delay, capacitance, DRC and EM all depend on metal rules.                         |
| 2 | Placement Optimization                      | Bad placement makes CTS and routing harder. Good placement reduces detours, capacitance and vio... |
| 3 | Clock Tree Synthesis                        | Clock arrival controls setup and hold timing. CTS changes ideal-clock timing into real-clock ti... |
| 4 | Routing and Post-Route Optimization         | Final timing and physical verification depend on routed wires, not estimates.                      |

## Metal Stack, Width, Spacing, Pitch and Vias

| Part                      | Explanation                                                                                                                                           |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| What it is                | Metal stack defines routing layers. Width is wire thickness, spacing is gap between wires, pitch is track-to-track distance, and vias connect layers. |
| Why it matters in ASIC/PD | Routing quality, delay, capacitance, DRC and EM all depend on metal rules.                                                                            |

## Examples

- Long nets use wider upper metals.
- Adjacent layers alternate horizontal and vertical direction.
- Multi-cut vias reduce resistance.

## Common Mistakes

- Using lower thin metals for long/high-current nets.
- Ignoring via enclosure and spacing rules.

## Practice Questions

- Define pitch.
- Why use via arrays?
- Why are upper metals better for long routes?

## Placement Optimization

| Part                      | Explanation                                                                                                    |
|---------------------------|----------------------------------------------------------------------------------------------------------------|
| What it is                | Placement assigns standard cells to legal locations while optimizing timing, congestion, power and wirelength. |
| Why it matters in ASIC/PD | Bad placement makes CTS and routing harder. Good placement reduces detours, capacitance and violations.        |

## Examples

- Global placement spreads cells.
- Legalization removes overlaps.
- Detailed placement improves local timing/congestion.

## Common Mistakes

- Ignoring congestion until route.
- Over-dense cells around macro pins.

## Practice Questions

- What is legalization?
- What is congestion overflow?
- How does placement affect setup timing?



## Clock Tree Synthesis

| Part                      | Explanation                                                                                             |
|---------------------------|---------------------------------------------------------------------------------------------------------|
| What it is                | CTS builds real clock networks using buffers/inverters and routing rules to deliver clock to all sinks. |
| Why it matters in ASIC/PD | Clock arrival controls setup and hold timing. CTS changes ideal-clock timing into real-clock timing.    |

## Examples

- Skew is difference between clock arrival times.
- Insertion delay is source-to-sink clock delay.
- Useful skew can help setup but hurt hold.

## Common Mistakes

- Ignoring generated clocks.
- Over-buffering clocks and increasing power.

## Practice Questions

- What is skew?
- What is insertion delay?
- Why do hold violations appear after CTS?

## Routing and Post-Route Optimization

| Part                      | Explanation                                                                                                                       |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| What it is                | Routing connects all nets with legal wires and vias. Post-route optimization fixes timing, SI and DRC using realistic parasitics. |
| Why it matters in ASIC/PD | Final timing and physical verification depend on routed wires, not estimates.                                                     |

### Examples

- Global route plans paths.
- Detailed route creates wires/vias.
- Antenna diode can fix plasma charging risk.

### Common Mistakes

- Leaving shorts/opens for signoff.
- Not rechecking timing after route ECO.

### Practice Questions

- What are shorts and opens?
- What is antenna violation?
- Why run extraction after route?

# EXPERT - Signoff, DRC/LVS, IR/EM, SI, ECO and Tapeout Readiness

This module teaches final closure: proving the routed design is timing-clean, manufacturable, reliable and ready for foundry handoff.

| # | Topic                         | Main Skill                                                                                         |
|---|-------------------------------|----------------------------------------------------------------------------------------------------|
| 1 | RC Extraction and Signoff STA | Estimated timing is not enough for tapeout. Final wire parasitics can change setup, hold and SI... |
| 2 | DRC, LVS, Antenna and Density | Foundry will not accept a layout that violates manufacturing or connectivity rules.                |
| 3 | IR/EM and Signal Integrity    | A timing-clean chip can still fail if voltage droops, wires degrade or coupling noise corrupts ... |
| 4 | ECO and Tapeout Package       | Late changes must be small, scripted and re-signed. Tapeout requires traceable files and report... |

## RC Extraction and Signoff STA

| Part                      | Explanation                                                                                                                                                      |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What it is                | Extraction converts routed metals and vias into parasitic resistance/capacitance. Signoff STA checks timing with final netlist, constraints, libraries and SPEF. |
| Why it matters in ASIC/PD | Estimated timing is not enough for tapeout. Final wire parasitics can change setup, hold and SI behavior.                                                        |

### Examples

- Cmax SPEF is setup-sensitive.
- Cmin SPEF is hold-sensitive.
- PrimeTime/Tempus reports setup and hold across scenarios.

### Common Mistakes

- Using pre-fill SPEF for final signoff.
- Leaving unconstrained endpoints.

### Practice Questions

- What is SPEF?
- Why use Cmax and Cmin?
- What does signoff STA prove?

## DRC, LVS, Antenna and Density

| Part                      | Explanation                                                                                                                                                           |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What it is                | DRC checks manufacturing rules. LVS checks layout connectivity against netlist. Antenna checks plasma charging risk. Density checks metal fill/manufacturing windows. |
| Why it matters in ASIC/PD | Foundry will not accept a layout that violates manufacturing or connectivity rules.                                                                                   |

### Examples

- Spacing violation is DRC.
- Missing connection can cause LVS mismatch.
- Metal fill helps meet density but changes capacitance.

### Common Mistakes

- Not re-running DRC/LVS after ECO.
- Treating all violations as waivable.

### Practice Questions

- Difference between DRC and LVS?
- Why add metal fill?
- What is antenna diode?

## IR/EM and Signal Integrity

| Part                      | Explanation                                                                                                                   |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| What it is                | IR drop checks voltage loss in power grid. EM checks current density reliability. SI checks crosstalk noise and delay impact. |
| Why it matters in ASIC/PD | A timing-clean chip can still fail if voltage droops, wires degrade or coupling noise corrupts signals.                       |

### Examples

- Dynamic IR occurs during high switching.
- EM risk is high on narrow high-current wires.
- Crosstalk can slow or speed neighboring signals.

### Common Mistakes

- Using unrealistic activity for power signoff.
- Ignoring clock and SRAM interface hotspots.

### Practice Questions

- What causes IR drop?
- What is EM?
- How does crosstalk affect timing?

## ECO and Tapeout Package

| Part                      | Explanation                                                                                                                                    |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| What it is                | ECO is a controlled engineering change after implementation. Tapeout package is the frozen release set sent for manufacturing/signoff archive. |
| Why it matters in ASIC/PD | Late changes must be small, scripted and re-signed. Tapeout requires traceable files and reports.                                              |

## Examples

- Timing ECO inserts buffers or resizes cells.
- Functional ECO changes logic using spare cells.
- Release package includes GDS/OASIS, DEF, Verilog, SPEF, SDF, SDC, reports and checksums.

## Common Mistakes

- Manual GUI edits with no script.
- No checksum or waiver documentation.

## Practice Questions

- Types of ECO?
- Why use spare cells?
- What files are in tapeout release?